Project Proposal

🎯 **Objective**:

The purpose of the project is to apply Python programming skills to solve a data science-related problem. Students will start by writing a proposal that outlines their project idea and presenting it for approval. This ensures clarity in objectives and approach before moving to the implementation phase.

"Be creative, but ensure your project involves working with data!"

✨ **Suggested Topics for Projects**

To help you brainstorm ideas for your project, here are some suggested topics. You may choose one of these or propose a unique idea of your own, as long as it aligns with the field of data science and utilizes Python.

1. **Student Grade Analyzer:**
   - A tool to manage and analyze student grades, allowing users to view, add, and calculate average grades for students.
   - Provides insights like identifying top-performing students and generating summary reports for academic performance.
2. **Expense Tracker:**
   - A tool to help users track their daily expenses by categorizing them (e.g., food, transport, entertainment).
   - Provides insights like total spending, savings, and generates a summary report for better financial management.
3. **Quiz Application:**
   - A console-based quiz app where users can take quizzes on various topics.
   - Tracks scores, provides feedback, and allows adding new questions for a dynamic quiz experience.
4. **Weather Data Analyzer:**
   - Analyzes weather data (e.g., temperature, humidity) from a dataset to identify trends and patterns.
   - Provides insights like average temperatures, hottest/coldest days, and weather extremes.

► **Note**: If you want to create your own project, ensure it aligns with data science concepts such as data analysis, visualization, or prediction. Here are some tips:

- Pick a real-world problem that involves working with data (e.g., sales, weather, health, or education).
- Your project should aim to extract meaningful insights or patterns from data.
- Start with a small dataset and basic analysis (e.g., averages, trends, or comparisons).
- (OPTIONAL) You can always add advanced features like visualizations (using matplotlib) or predictions (using basic machine learning).

► **Tip to Find a dataset**:

- **Use Popular Platforms**: Check websites like Kaggle, UCI Machine Learning Repository, or Google Dataset Search.
- **Explore Government Data**: Visit portals like data.gov or World Bank Open Data for public datasets.
- **Search GitHub**: Look for repositories with "awesome datasets" or use Awesome Public Datasets.
- **Check Reddit**: Visit r/datasets for unique or niche datasets shared by the community.
- **Focus on Relevance**: Choose datasets that match your project topic and are easy to work with (e.g., CSV or JSON formats).

🤝 **Requirements**

Your project must fulfill the following mandatory requirements:

1. **Use of Data Structures in Python**:
   - Incorporate at least one data structure (list, tuple, dictionary, or set) in your project.
2. **Implementation of OOP Concepts**:
   - Your project should demonstrate the use of classes and objects, along with **at least two** of the following OOP principles:
     i. Encapsulation
     ii. Inheritance
     iii. Polymorphism

iv. Abstraction

3. **Magic Methods**
   - Include at least five different magic methods in your project.
4. **File Handling**:
   - Include file handling for tasks like reading/writing data, logging events, or managing input/output operations.
5. **Error Handling**:
   - Use try-except blocks to handle potential errors gracefully, ensuring your program doesn't crash unexpectedly.

🔆 Part 1: Writing the Project Proposal

Your proposal should clearly outline the following sections:

1. **Project Title:**
   - A concise and descriptive title that reflects the essence of your project.
2. **Introduction:**
   - Briefly describe the context or problem your project addresses.
   - Why is this project important in the field of data science?
3. **Objectives:**
   - Clearly state what you aim to achieve with your project.
   - Objectives should be measurable and specific.
4. **Proposed Solution:**
   - Explain how you plan to address the problem using Python.
   - Include key features and functionalities you intend to implement.
   - Highlight how Python concepts like OOP, file handling, or error handling will be utilized.
5. **Methodology:**
   - Outline the steps you will take to develop the project.
   - Include tools, libraries, and frameworks you will use (e.g., hashlib, os, math).
   - Describe how you will test the security features of your project.
6. **Timeline:**
   - Provide a week-by-week breakdown of your project plan (*3 weeks in total, counting from 23rd February 2026*).
   - Specify milestones, such as completing specific modules or features.
7. **Expected Outcome:**
   - Describe what the final output of your project will look like.

- Mention its potential impact or use in data science.

8. **References (if any):**
   - Include any sources or research you consulted while drafting your proposal.

💢 **Formatting:**

- Use clear headings for each section.
- Ensure your proposal is 1-2 pages long and professionally formatted.

🔆 **Part 2: Proposal Presentation**

Prepare a 7-10 minute presentation to present your proposal. Focus on:

1. **Overview of the Project:**
   - Introduce your project title and the problem it solves.
   - Briefly explain why this project is significant.
2. **Key Features:**
   - Highlight the main features of your solution.
   - Mention how you will incorporate Python concepts and tools.
3. **Feasibility:**
   - Justify why your proposed solution is achievable within the given timeline and your skill set.
4. **Timeline Overview:**
   - Share a summarized version of your project timeline and milestones.

💢 **Deadline of submission:**

☺ Only **ONE member** from each team should submit the completed documents needed by Saturday, 28th February 2026, at 5 PM.

☺ There are 3 files in one zip file to submit:

[Group_number]_[Team_number]_Proposal.zip

⇒ [Group_number]_[Team_number]_Proposal.pdf

- e.g., G1_T8_Proposal.pdf

⇒ [Group_number]_[Team_number]_Slide_Proposal.pptx

⇒ [Group_number]_[Team_number]_Slide_Proposal.pdf